

A) 3 Stages:

Machine Learning – As the data contains structural(rows, columns) numeric & text values

Supervised – As the requirements are clear, Input and output labels are exists in the data.

Regression – As the output variable charges contains continuous values.

B) Problem Statement: Predict the insurance charges

C) Algorithms:

1) **LinearRegression:**

R2 score: 0.768

2) **SVM:**

Class	Kernel	Gamma	R2_Score
SVR	Rbf(default)	Scale(default)	-0.0816
SVR	Rbf(default)	auto	-0.0829
SVR	linear	auto	-0.1005
SVR	linear	Scale(default)	-0.1005
SVR	poly	Scale(default)	-0.0642
SVR	poly	auto	Kernel restart
SVR	sigmoid	auto	Kernel restart
SVR	sigmoid	Scale(default)	-0.0829
SVR	precomputed	Scale(default)	Must be square matrix. Here input is 35*5
NuSVR	Rbf(default)	Scale(default)	-0.0233
NuSVR	Rbf(default)	auto	-0.0264
NuSVR	linear	auto	0.1152
NuSVR	linear	Scale(default)	Kernel restart
NuSVR	poly	Scale(default)	0.0119
NuSVR	poly	auto	Kernel restart
NuSVR	sigmoid	auto	Kernel restart
NuSVR	sigmoid	Scale(default)	-0.0270
NuSVR	precomputed	Scale(default)	Must be square matrix. Here input is 35*5

Class	penalty	loss	R2_Score
LinearSVR	-	epsilon_insensitive(default)	-0.1093
LinearSVR	-	squared_epsilon_insensitive	0.7558

From the above, Best R2 score value is 0.7558. Class: LinearSVR, loss= squared_epsilon_insensitive.

Decision Tree:

Class	criterion	splitter	R2_Score
DecisionTreeRegressor	Squared_error(default)	best(default)	0.7586
DecisionTreeRegressor	Squared_error(default)	random	0.7108
DecisionTreeRegressor	Friedman_mse	best(default)	0.7736
DecisionTreeRegressor	Friedman_mse	random	0.7288
DecisionTreeRegressor	Absolute_error	best(default)	0.7695
DecisionTreeRegressor	Absolute_error	random	0.6867
DecisionTreeRegressor	poisson	best(default)	0.7931
DecisionTreeRegressor	poisson	random	0.7283
ExtraTreeRegressor	Squared_error(default)	random (default)	0.7007
ExtraTreeRegressor	Squared_error(default)	best	0.7512
ExtraTreeRegressor	Friedman_mse	random (default)	0.6865
ExtraTreeRegressor	Friedman_mse	best	0.7664
ExtraTreeRegressor	Absolute_error	random (default)	0.7344
ExtraTreeRegressor	Absolute_error	best	0.7433
ExtraTreeRegressor	poisson	random (default)	0.7149
ExtraTreeRegressor	poisson	best	0.7848

From the above, Best R2 score value is 0.7931. Class: DecisionTreeRegressor, criterion: poisson, splitter = best.

Random Forest:

Class	N_estimators	criterion	Max_depth	R2_Score
RandomForestRegressor	100(default)	Squared_error(default)	None(default)	0.8439
RandomForestRegressor	100(default)	Squared_error(default)	10	0.8416
RandomForestRegressor	100(default)	absolute_error	None(default)	0.8330
RandomForestRegressor	100(default)	absolute_error	10	0.8484
RandomForestRegressor	100(default)	poisson	None(default)	0.8399
RandomForestRegressor	100(default)	poisson	10	0.8421
RandomForestRegressor	1000	Squared_error(default)	None(default)	0.8403
RandomForestRegressor	1000	Squared_error(default)	10	0.8432
RandomForestRegressor	1000	absolute_error	None(default)	0.8353
RandomForestRegressor	1000	absolute_error	10	0.8489
RandomForestRegressor	1000	poisson	None(default)	0.8421
RandomForestRegressor	1000	poisson	10	0.8455

From the above, Best R2 score value is 0.8489. Class: RandomForestRegressor, N_estimators: 1000, criterion: absolute_error, max_depth = 10.

Best R2 value from the above algorithms sorted in the order of highest value.

Algorithm	R2 Value	Hypertuning parameter
Random Forest	0.8489	Class: RandomForestRegressor, N_estimators: 1000, criterion: absolute_error, max_depth = 10.
Decision Tree	0.7931	Class: DecisionTreeRegressor, criterion: poisson, splitter = best.
Linear Regression	0.768	NA
SVM	0.7558	Class: LinearSVR, loss= squared_epsilon_insensitive.

For the Insurance dataset best datamodel is Random Forest algorithm with the hypertuning parameters. Class: RandomForestRegressor, N_estimators: 1000, criterion: absolute_error, max_depth = 10.